

INFO6007

Project Management in IT

## Week 05: Quality Management Plan

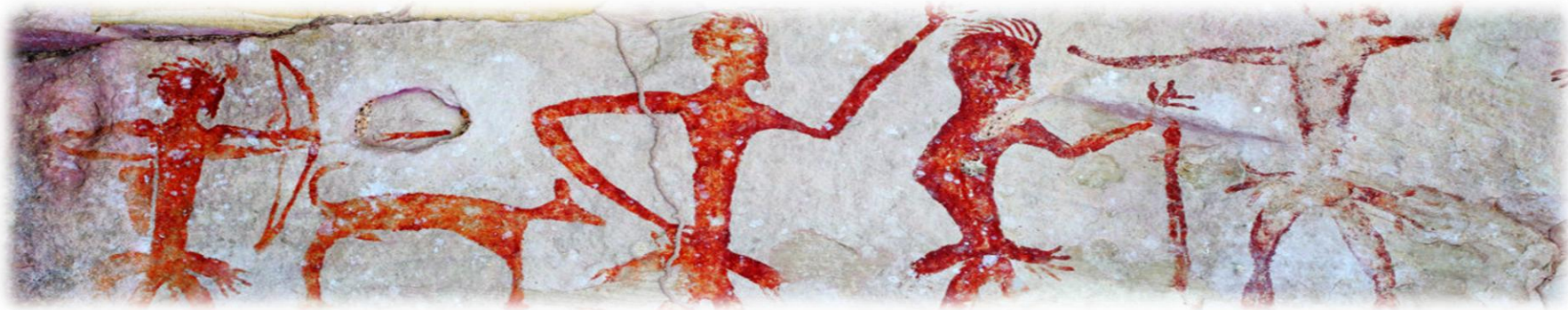
Semester 1, 2026



# Acknowledgement of Country

*I would like to acknowledge the Traditional Owners of Australia and recognise their continuing connection to land, water and culture. I am currently on the land of the Gadigal people of the Eora nation and pay my respects to their Elders, past, present and emerging.*

*I further acknowledge the Traditional Owners of the country on which you are on and pay respects to their Elders, past, present and future.*



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# Agenda

- Introduction
- Project Quality Management
- Tools and Techniques used in QMP
- Other important concepts



# Project Quality Management

# Introduction

- Quality is the degree to which a product or service meets requirements, expectations, and is fit for its intended purpose.
- The PMBOK defines Quality as: *"The degree to which a set of inherent characteristics fulfill requirements."*
- There are multiple dimensions of Quality in IT Projects:
  - **Functionality:** does it do what it's supposed to?
  - **Reliability:** does it work consistently without failure?
  - **Performance:** is it fast and efficient?
  - **Usability:** is it intuitive for end users?
  - **Maintenance:** can it be updated and improved easily?
  - **Security:** are data and systems protected?

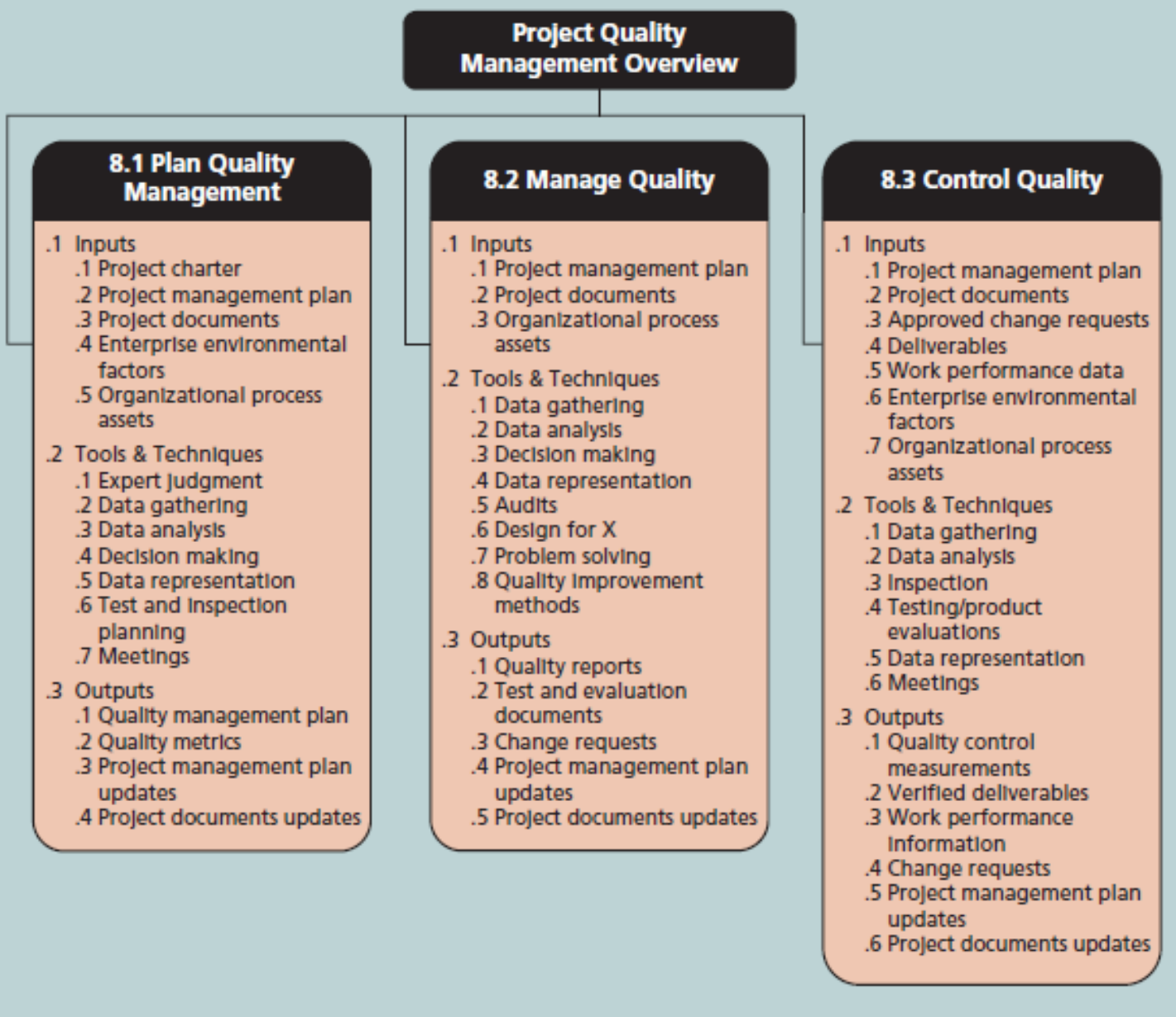
# Introduction

- There are different perspectives to Quality as well:
  - Conformance to Requirements
    - Did the project deliver what was documented in the scope/specifications?
    - Example: If a client asked for a payroll system, does it calculate salaries and taxes correctly?
  - Fitness for Use
    - Does the system actually serve the user effectively?
    - Example: A payroll system might “work” but if it’s too slow or difficult to use, it lacks quality.
  - Customer Satisfaction
    - Quality is ultimately judged by the end-user or client’s experience.
    - Example: Even bug-free software is “poor quality” if it fails to meet real business needs.

# Project Quality Management

- Project Quality Management ensures that a project's deliverables meet stakeholder expectations, requirements, and agreed standards, while also ensuring the project processes themselves are efficient and effective.
- It is about both:
  - Managing quality of the deliverables (product quality).
  - Managing quality of the project processes (process quality).
- Project Quality Management revolves around 4 major principles:
  - Customer focus: End-user satisfaction is the ultimate quality measure.
  - Prevention over Inspection: It's cheaper to prevent defects than fix them later.
  - Continuous Improvement: Using lessons learned and feedback loops (Agile, Kaizen, DevOps).
  - Fact-based decision making: Using metrics, dashboards, and data analysis.





## 3 main processes in Project Quality Management

Source: *PMBOK® Guide – Sixth Edition*. Project Management Institute, Inc. (2017). Copyright and all rights reserved. Material from this publication has been reproduced with permission of PMI.

**FIGURE 8-1** Project quality management overview

# 1. Planning Quality Management

- This is the process of identifying:
  - The **quality requirements** and **standards** for the project and deliverables.
  - How compliance will be demonstrated and measured.
  - The activities, resources, and metrics needed to ensure quality is built in from the start.

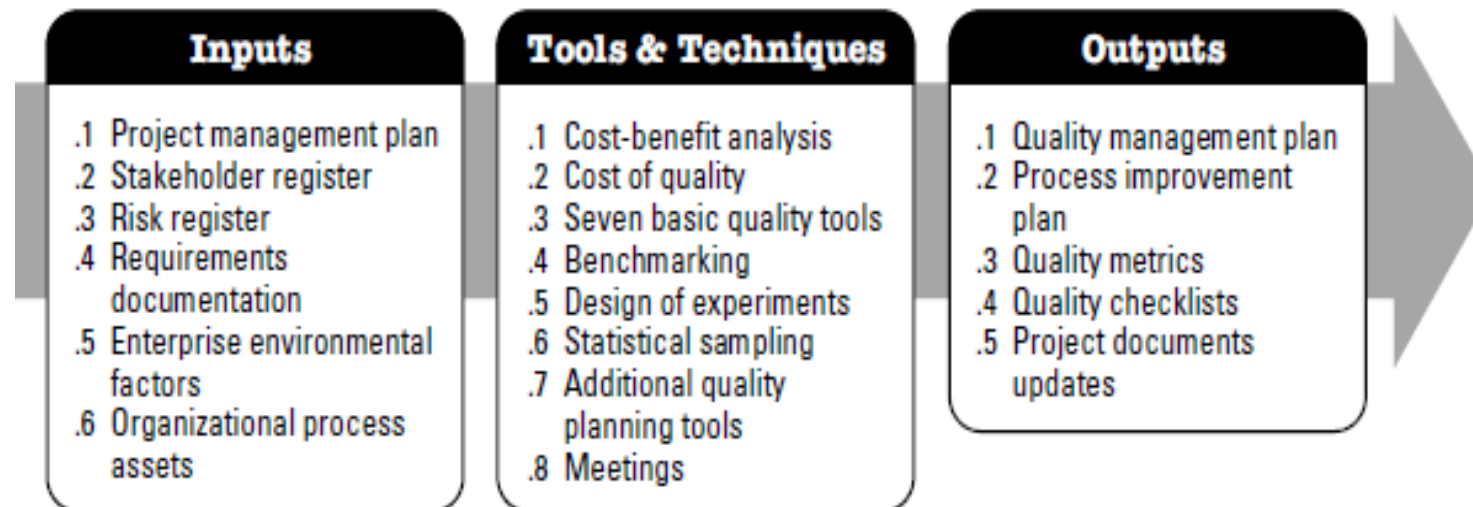


Figure 8-3. Plan Quality Management Inputs, Tools & Techniques, and Outputs

# 1. Planning Quality Management

- The main objectives of this step is to:
  - Define what “quality” means for this project.
  - Identify standards, policies, and metrics to use.
  - Determine processes, tools, and techniques to achieve quality.
  - Clarify roles & responsibilities (who checks what, when, and how).
- Key activities in Planning include:
  - Identifying Quality Standards
  - Defining Quality Metrics
  - Planning Quality Assurance Activities
  - Planning Quality Control Activities
  - Documenting in the Quality Management Plan

# Roles and Responsibilities in QMP

- Project Manager
  - Develops and maintains the Quality Management Plan.
  - Ensures quality standards align with project objectives and client requirements.
  - Balances trade-offs between cost, time, and quality.
  - Monitors CoQ, CoC, and CoNC.
  - Approves corrective and preventive actions.
  - Communicates quality requirements to stakeholders.
- Quality Assurance Lead/QA Team
  - Define and implement QA processes, methods, and metrics.
  - Conduct audits to verify adherence to standards.
  - Develop test strategies and ensure compliance with organizational quality policies.
  - Report quality issues and risks to the PM.
  - Facilitate continuous improvement initiatives.

# Roles and Responsibilities in QMP

- Quality Control/Testing team
  - Execute tests (unit, integration, system, UAT, performance, etc.).
  - Document defects, errors, and non-conformities.
  - Use quality tools (checklists, control charts, Pareto, etc.) to monitor quality.
  - Validate fixes and verify deliverables meet requirements.
  - Provide feedback to development teams for defect prevention.
- Development Team
  - Follow coding standards, guidelines, and defined processes.
  - Perform peer reviews and unit testing before handover to QA.
  - Implement preventive and corrective actions.
  - Collaborate with QA/QC teams to resolve issues.
  - Maintain documentation for traceability.



# Quality Standards in IT

- ISO 9001 – Quality management Systems
  - General-purpose standard for implementing QMS in any industry, including IT.
  - Focuses on continuous improvement, customer satisfaction, and documented processes.



(ISO:9001 Information, 2022)

- ISO/IEC 27001 – Information Security Management
  - Security as a core part of IT quality.
  - Ensures confidentiality, integrity, and availability of data.
  - Often required for financial services, healthcare, and government IT projects.



(ISO:27001 Information, 2022)

# Quality Standards in IT

- Capability Maturity Model Integration
  - This is a framework for assessing and improving software development processes.
  - Constitutes of 5 levels:
    - Level 1: Initial - unpredictable, ad hoc.
    - Level 2: Managed - processes are project-specific.
    - Level 3: Defined - processes are standardized across the org.
    - Level 4: Quantitatively Managed - metrics driven processes.
    - Level 5: Optimizing - continuous improvement.



# Quality Standards in IT

- Information Technology Infrastructure Library (ITIL)
  - Framework for IT Service Management (ITSM).
  - Quality defined as: ensuring services are aligned with business needs and provide value.
  - Emphasis on Service Level Agreements (SLAs) and continuous service improvement.
  - We will learn more about this specific framework in the future.



## ITIL Framework

(Thakur, n.d.)

## 2. Managing Quality

- Managing Quality is the process of **translating the Quality Management Plan into executable quality activities**.
  - It ensures that the project processes are being followed.
  - Focuses on preventing defects rather than just detecting them.
  - In IT projects, this means embedding quality into development, testing, and delivery workflows.
- Key activities in this step are:
  - Process Audits
  - Quality Assurance Reviews
  - Continuous Process improvements
  - Risk and Defect prevention

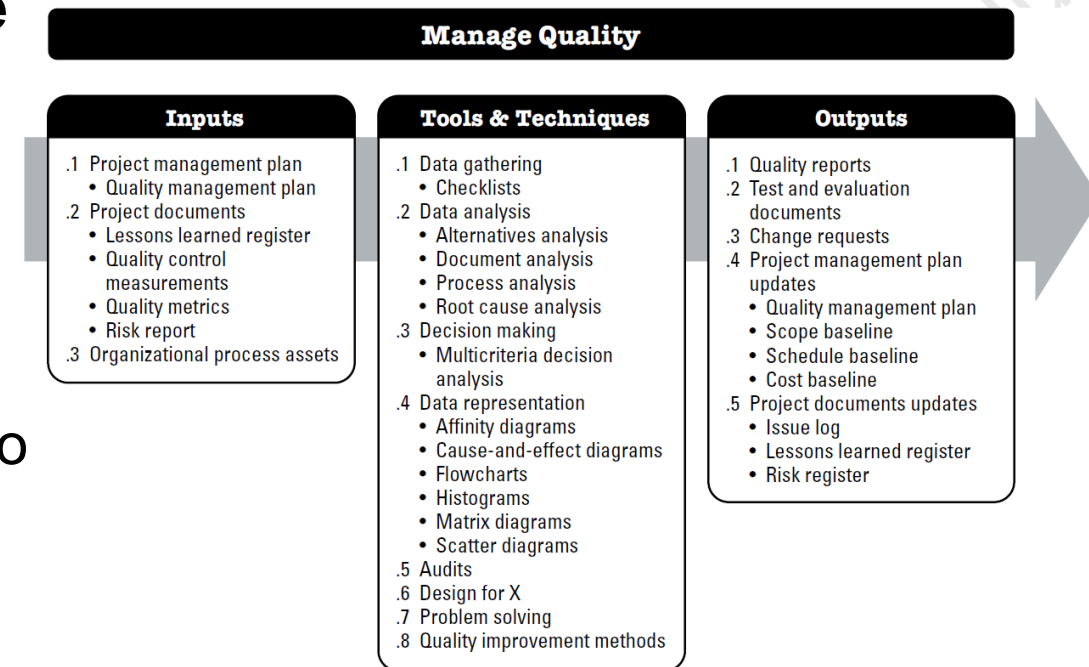


Figure 8-7. Manage Quality: Inputs, Tools & Techniques, and Outputs

# Process Audits

- It is a systematic review of project processes to check if teams are following defined quality standards, policies, and procedures.
- It is an important activity as Audits ensure compliance (e.g., ISO, CMMI, Agile standards) and catch process issues before they lead to product defects.
- Example:
  - Auditing whether developers are using secure coding practices (e.g., OWASP).
  - Checking if Agile teams are updating user stories with acceptance criteria.
  - Verifying CI/CD pipelines are properly logging test results.

# Quality Assurance Reviews

- It is the process of structured evaluations of deliverables and processes, performed throughout the project. Includes peer reviews, design walkthroughs, and retrospectives.
- This activity promotes collaboration, catches issues early, and improves knowledge sharing.
- Example:
  - Peer review of code to ensure maintainability and adherence to coding standards.
  - Architecture review session to evaluate scalability of a microservices design.
  - Agile sprint retrospective to identify process gaps and improve teamwork.



# Continuous Process Improvements

- It is an ongoing effort to refine processes for better efficiency, effectiveness, and adaptability. Often guided by **PDCA (Plan–Do–Check–Act)** or **Six Sigma DMAIC**.
- This activity reduces technical debt, eliminates bottlenecks, and builds a culture of learning.
- Example:
  - Improving build automation to reduce manual deployment errors.
  - Using metrics (e.g., bug turnaround time) to optimize QA workflows.
  - Adopting new tools (e.g., migrating from manual regression testing → automated regression suite).

# Agile and DevOps perspective of QMP

- Agile based QMP
  - Shared Responsibility for Quality
  - Customer-Centric Quality
  - Continuous Testing
  - Frequent Feedback
  - Adaptive QMP
- DevOps based QMP
  - Automation Everywhere
  - Continuous Integration & Continuous Delivery (CI/CD)
  - Monitoring & Observability
  - Infrastructure as Code (IaC)

Focuses on delivering quality incrementally.

Focuses on ensuring continuous quality from end-to-end.

# Risk and Defect prevention

- It is the process of identifying potential process weaknesses or risks that could create defects, and addressing them before they occur.
- This is a very important activity as prevention is far cheaper than fixing defects later (cost of quality principle).
- Example:
  - Using static code analysis tools (e.g., SonarQube) to prevent vulnerabilities before release.
  - Conducting threat modeling sessions to anticipate cybersecurity risks.
  - Reviewing third-party API dependencies to reduce risk of future integration failures.

# Menti – Quality Management for your project...

- <https://www.menti.com/ala2y6fvdcwz>



### 3. Controlling Quality

- Controlling Quality (or **Quality Control, QC**) is the process of **monitoring and verifying that deliverables meet defined quality standards and requirements**.
- This step is more reactive → detects defects in deliverables.
- Focuses on **outputs** (products, services, results), not processes.
- Ensures only **approved deliverables** are accepted.

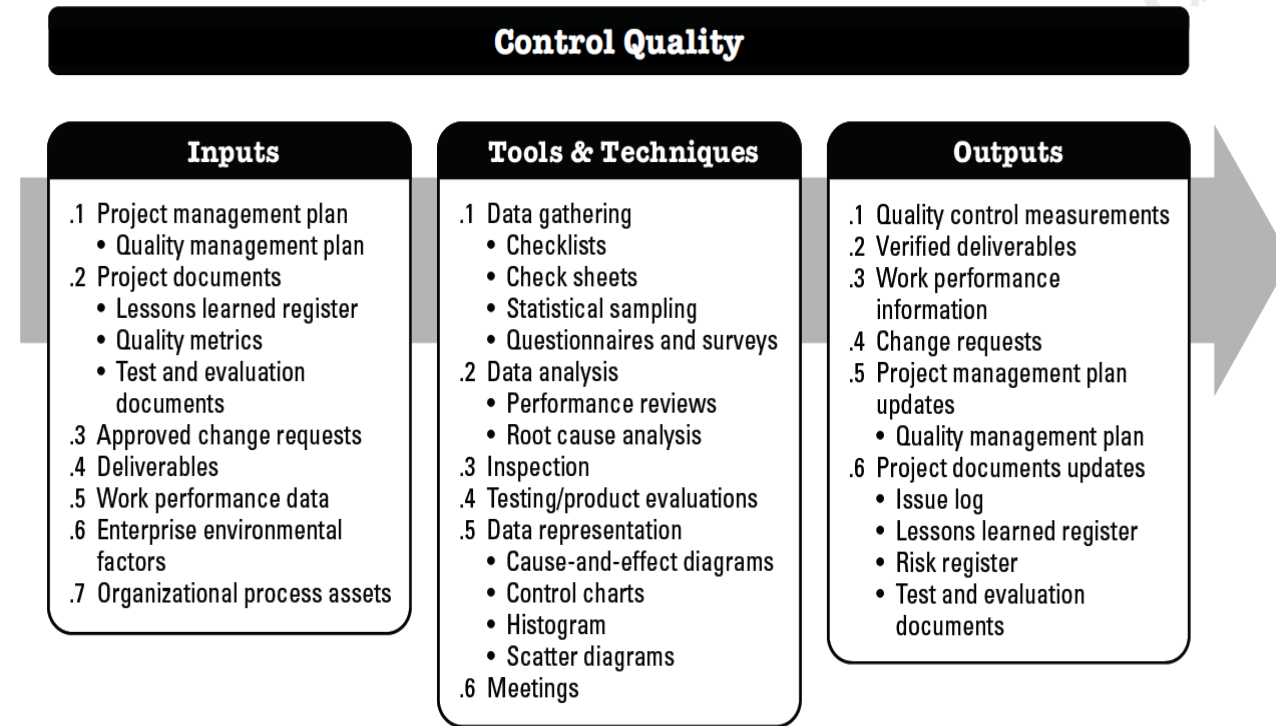


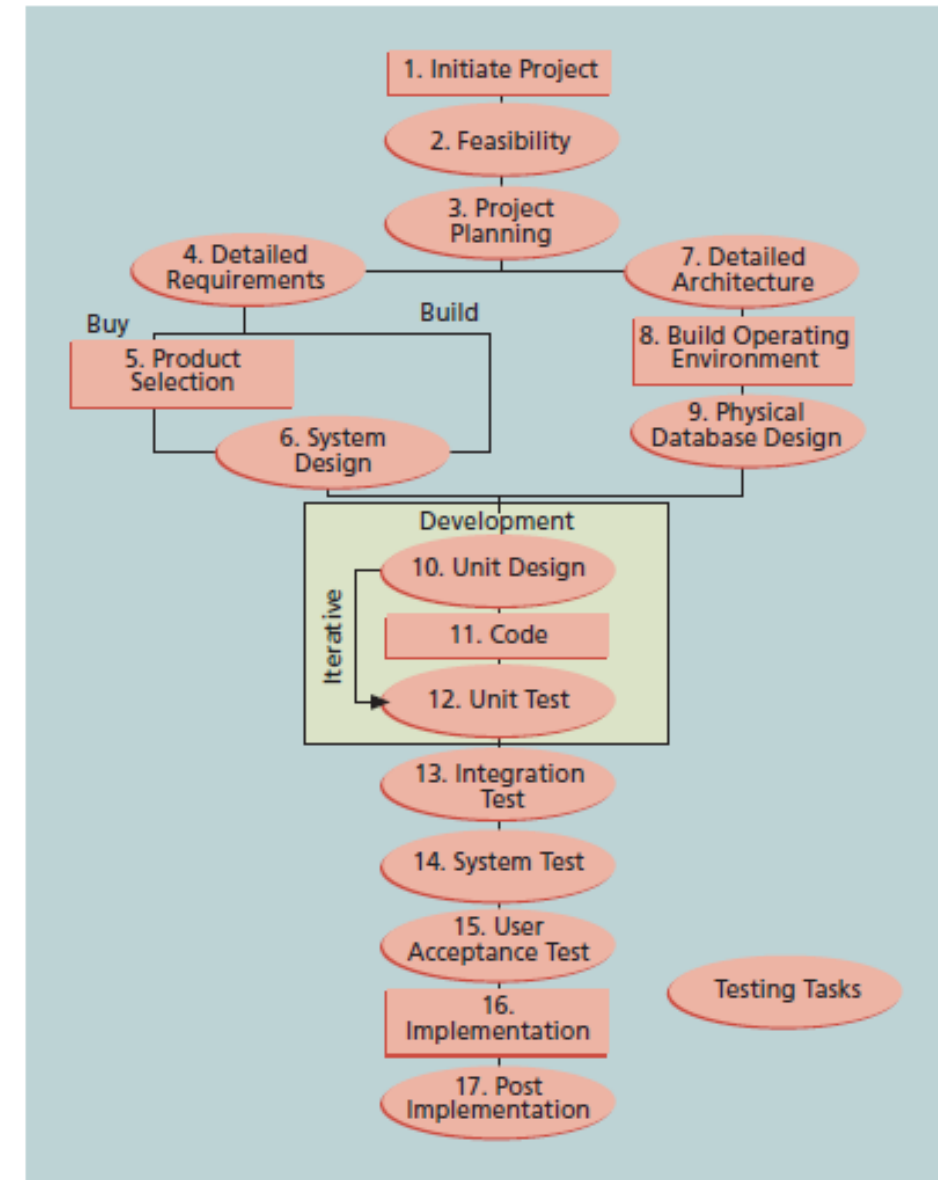
Figure 8-10. Control Quality: Inputs, Tools & Techniques, and Outputs

### 3. Controlling Quality

- Key activities in Quality control include:
  - Testing and Inspection
    - Verifying deliverables against defined requirements.
  - Defect identification and logging
    - Recording issues found during testing.
  - Measurement against Quality metrics
    - Comparing actual performance with planned thresholds.
  - Acceptance decisions
    - Deliverables are either **accepted, conditionally accepted, or rejected**.
  - Verification and Validation
    - **Verification:** "Did we build it right?" (meets specs).
    - **Validation:** "Did we build the right thing?" (meets user needs).

# Types of Testing for Quality Control

- Unit testing
  - Tests individual components or modules of the software in isolation.
- Integration testing
  - Tests the interaction between multiple modules or components.
- System testing
  - Tests the complete integrated system against requirements.
- User Acceptance testing
  - Final testing performed by end-users or clients.
- And many more!



Source: Hollstadt & Associates, Inc.

FIGURE 8-11 Testing tasks in the software development life cycle

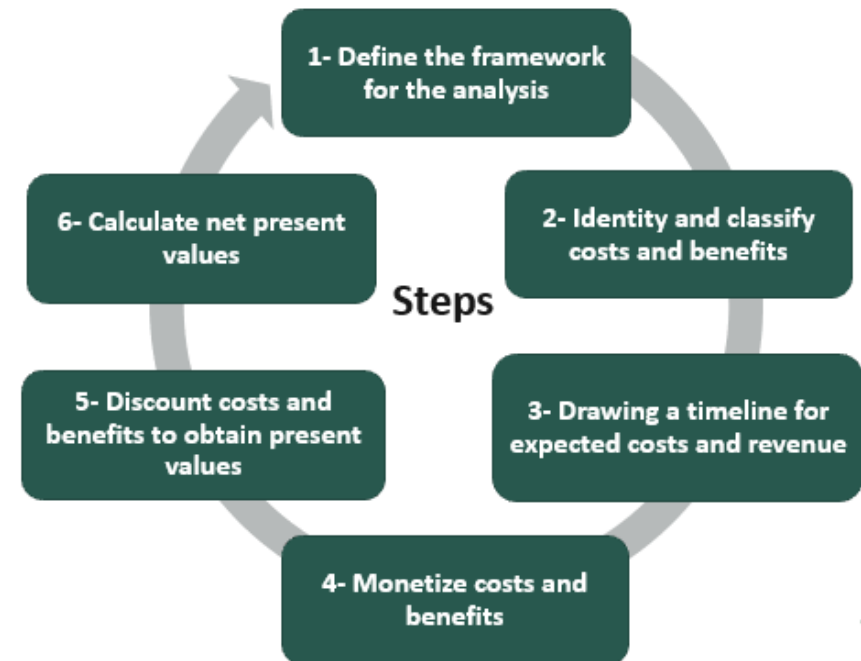


# Tools and Techniques used in Quality Management Plan

# Cost-Benefit Analysis

- This technique evaluates the cost of implementing a quality activity against the expected benefits.
- It helps prioritize quality initiatives based on Return On Investment (ROI).
- Example: Deciding whether to implement automated testing
  - Cost of automated testing tools vs. reduced defects and faster release cycles.

## Cost-Benefit Analysis



(Cost Benefit Analysis, 2024)

# Cause and Effect Diagram

- It is a visual tool to identify potential root causes of quality problems.
- It helps teams systematically investigate defects or issues.

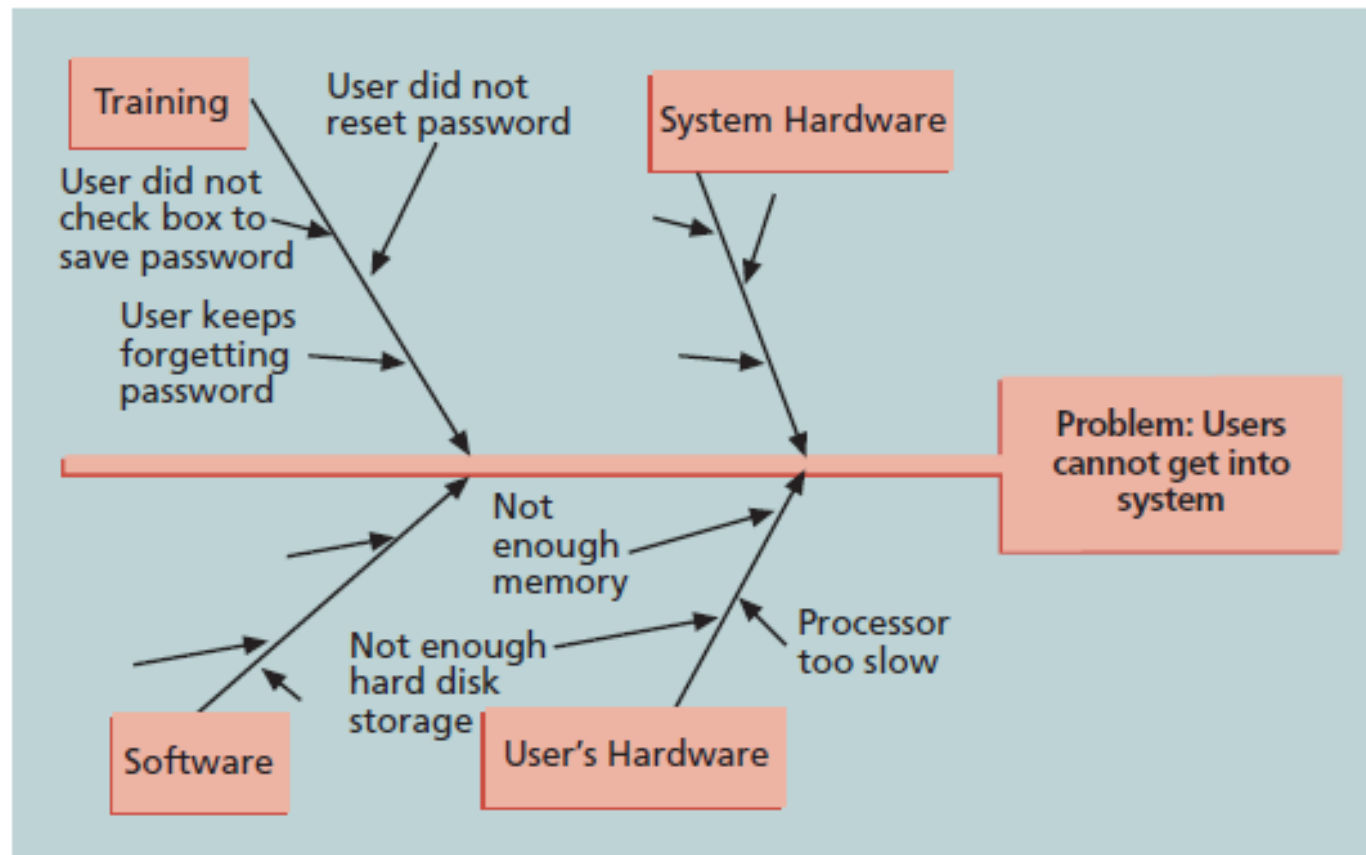


FIGURE 8-2 Sample cause-and-effect diagram

# Quality Control Chart

- It is a graphical representation showing process performance over time with upper and lower control limits.
- It detects variations or trends outside acceptable thresholds.

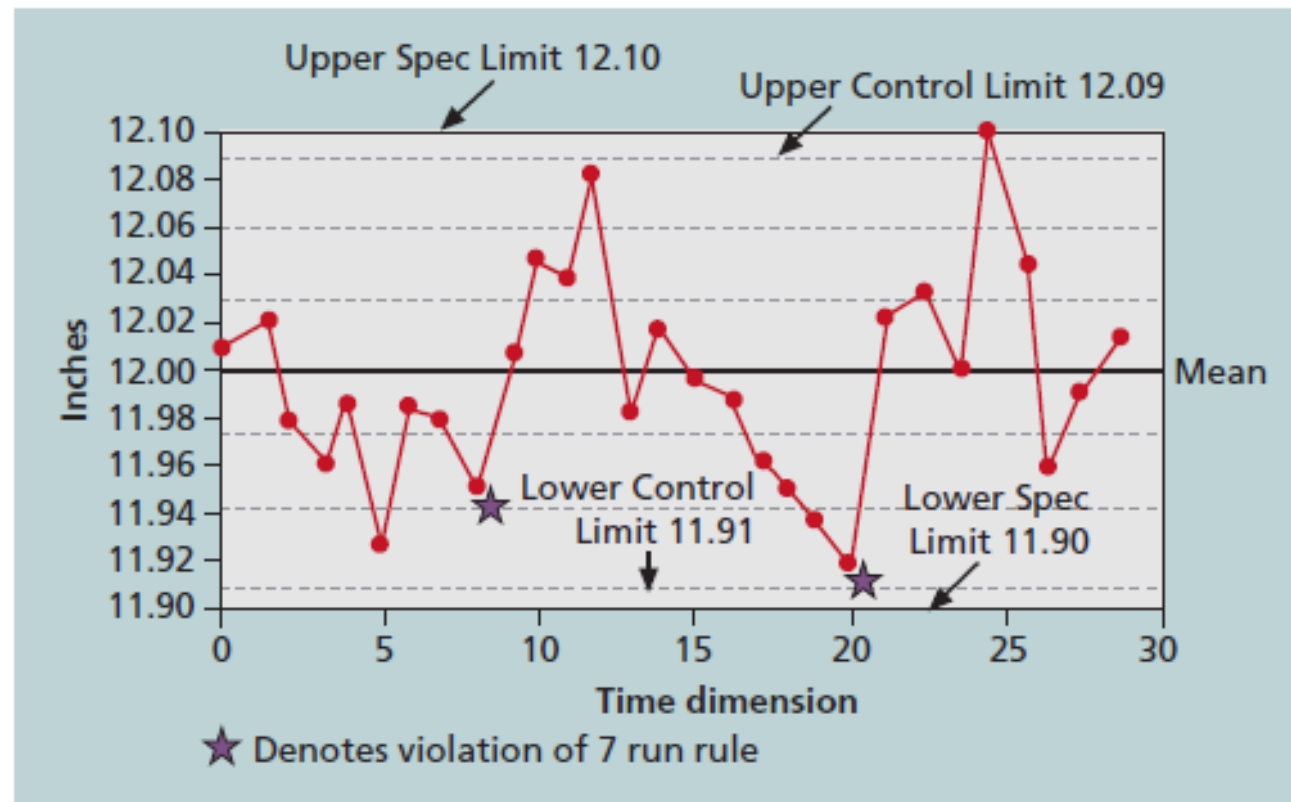


FIGURE 8-3 Sample control chart

# Checksheet

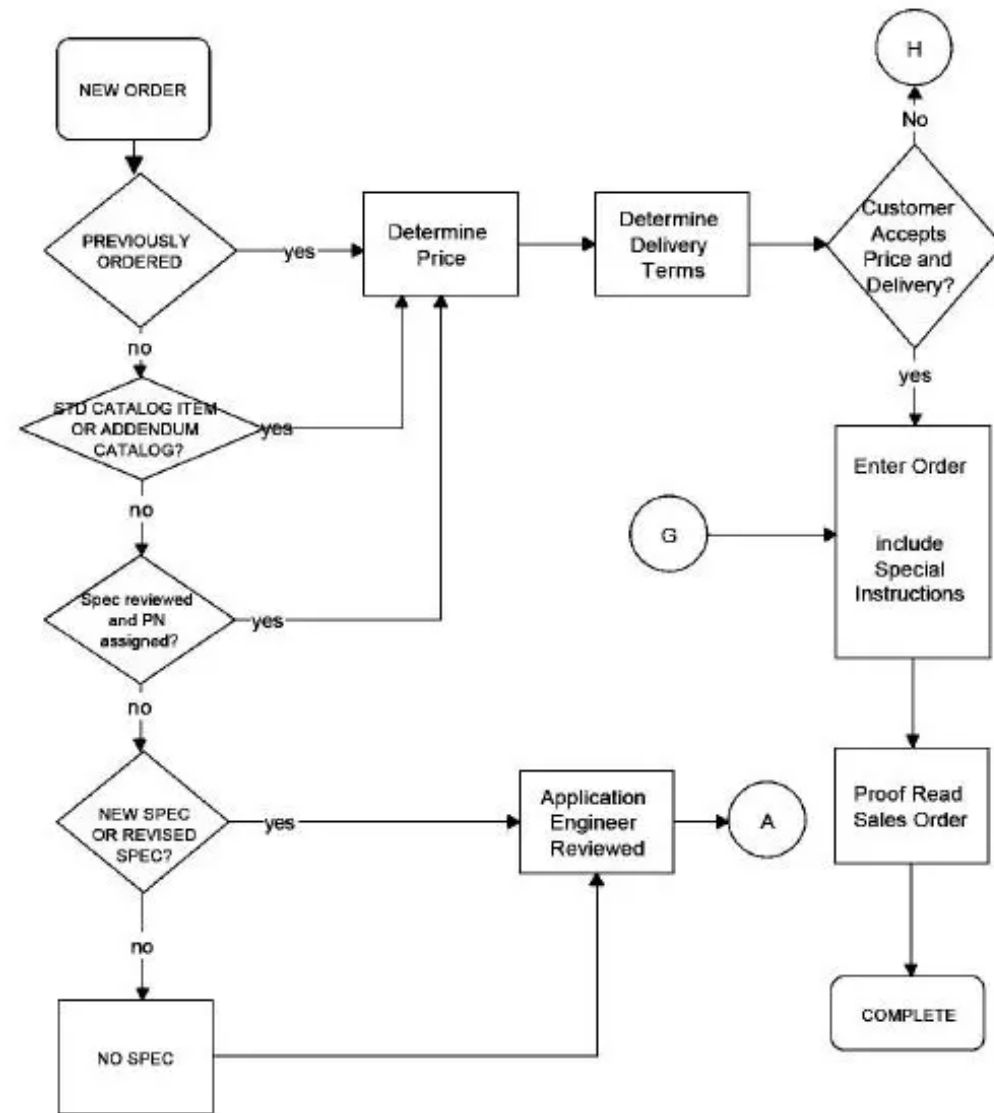
- It is a structured form used to collect and tally data in real time.
- It simplifies data collection for defects, errors, or process failures.

| System Complaints |        |         |           |          |        |          |        |       |
|-------------------|--------|---------|-----------|----------|--------|----------|--------|-------|
| Source            | Day    |         |           |          |        |          |        |       |
|                   | Monday | Tuesday | Wednesday | Thursday | Friday | Saturday | Sunday | Total |
| E-mail            |        |         |           |          |        |          |        | 12    |
| Text              |        |         |           |          |        |          |        | 29    |
| Phone call        |        |         |           |          |        |          |        | 8     |
| Total             | 11     | 10      | 8         | 6        | 7      | 3        | 4      | 49    |

FIGURE 8-4 Sample checksheet

# Flowcharts

- These are diagrams that map process steps visually.
- The main purpose of these diagrams are to identify process gaps, redundancies, and potential quality risk points.



(Broughton, n.d.)

Customer Service Responsibilities

# Scatter Diagram

- In the simplest sense, a scatter diagram is a graph showing relationship between two variables to identify correlation.
- The main purpose of drawing a scatter plot in Quality management is to detect potential causes of quality issues or trends.

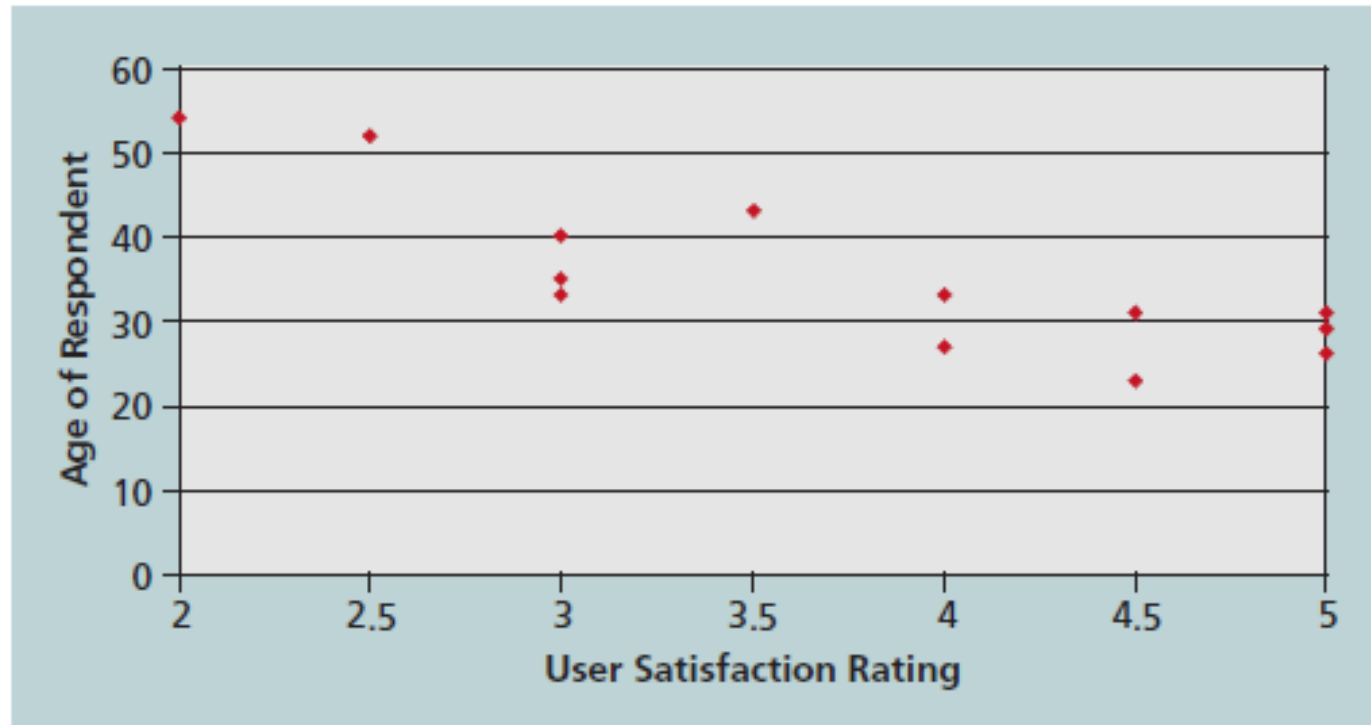
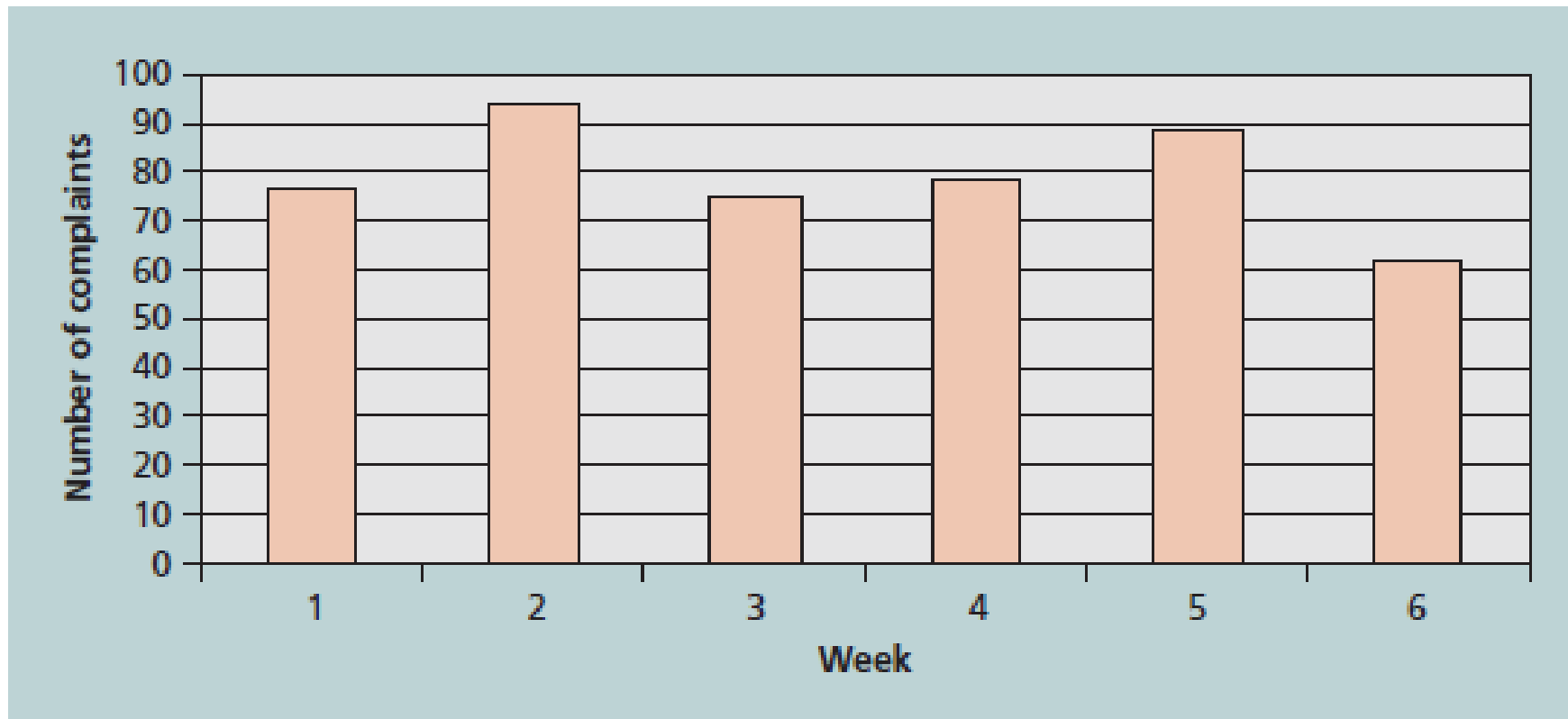


FIGURE 8-5 Sample scatter diagram

# Histogram

- It is a Bar chart showing frequency distribution of data.
- It helps understand patterns, common defect types, or error distribution.



**FIGURE 8-6** Sample histogram



# Pareto Chart

- It is a Bar chart combined with cumulative line graph to identify the most significant issues (80/20 rule).
- The **80/20 Rule** states that **roughly 80% of problems are caused by 20% of the causes**.
- It focuses improvement efforts on the problems causing the majority of defects.

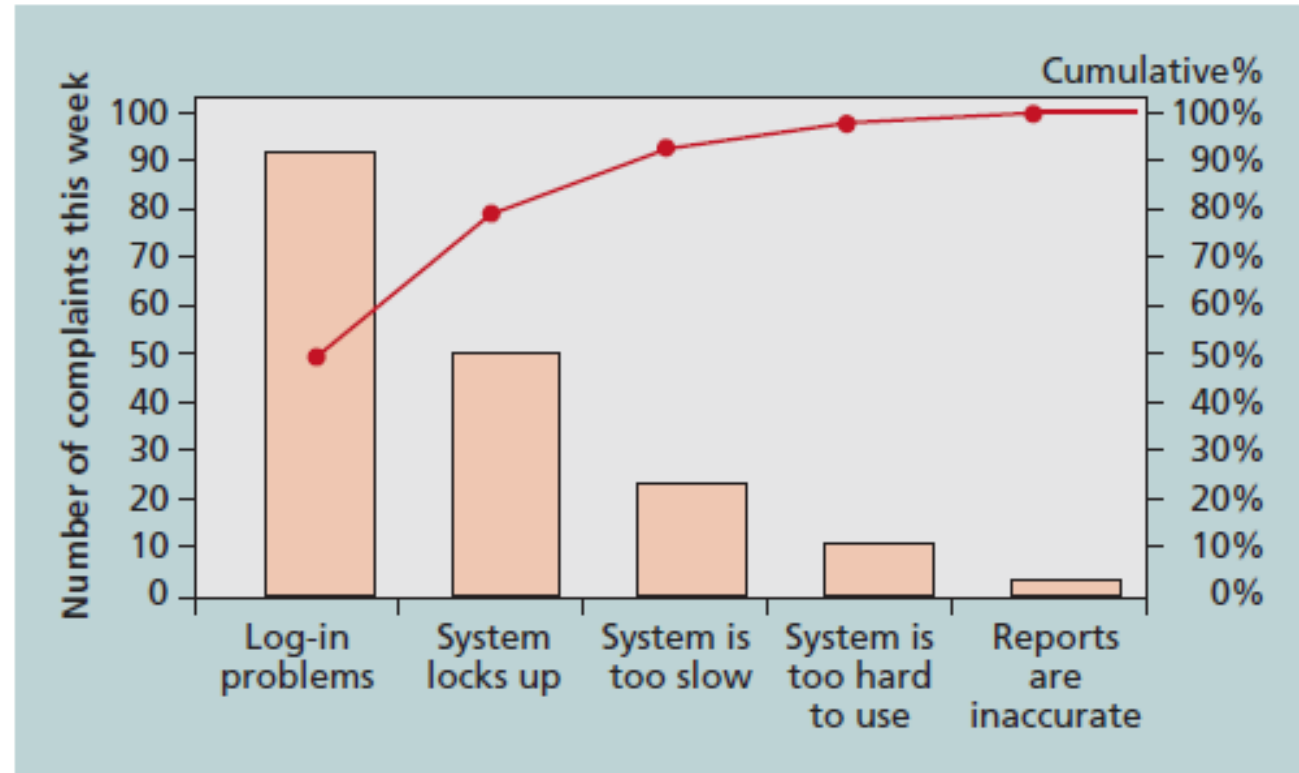


FIGURE 8-7 Sample Pareto chart

# Other Important Concepts

# Evolving IT Quality Practices

- As IT projects become more complex and fast-paced, traditional quality management approaches (checklists, inspections, late-stage testing) are no longer sufficient.
- Organisations now adopt **modern, evolving practices** that emphasize **speed, automation, and continuous improvement**.

# Lean Six Sigma in IT

- **Lean:** Focuses on eliminating waste (e.g., redundant processes, delays in defect resolution, excessive documentation).
- **Six Sigma:** Focuses on reducing variation and defects using data-driven approaches (DMAIC – Define, Measure, Analyze, Improve, Control).
- Application in IT Projects:
  - Streamlining development processes by reducing handoffs.
  - Applying defect root-cause analysis and statistical methods to improve software reliability.

# AI in Quality

- AI helps **predict and prevent quality issues** before they occur.
- Due to this, shifts quality assurance from reactive defect detection to proactive defect prevention.
- Example:
  - AI-driven code analysis to flag potential defects (e.g., SonarQube with ML).
  - Predictive analytics for project risk and defect probability.
  - AI-powered bots for exploratory testing and regression testing.

# Automation and DevOps Integration

- **Automated Testing:** Unit, integration, regression, and performance testing embedded in CI/CD pipelines.
- **Continuous Quality Gates:** Builds are automatically rejected if defects exceed thresholds.
- **Infrastructure Automation (IaC):** Prevents environment-related defects.
- **Benefits:** Faster releases, reduced human error, consistent quality across environments.

# Deming and his 14 points for management

1. Create constancy for purpose
  - Focus on long-term goals, quality improvement, and innovation, not just short-term profits.
2. Adopt the new philosophy
  - Embrace quality as everyone's responsibility.
3. Cease dependence on inspection to achieve quality
  - Build quality into processes rather than relying on post-production inspection.
4. End the practice of awarding business based on price alone
  - Focus on long-term supplier relationships and quality.

# Deming and his 14 points for management

## 5. Improve constantly and forever

- Continuous improvement of processes, products, and services.

## 6. Institute training on the job

- Ensure employees have the knowledge and skills needed for quality.

## 7. Adopt and institute leadership

- Leaders should help people do a better job, not just supervise.

## 8. Drive out fear

- Encourage open communication and reporting of problems without fear.

## 9. Break down barriers between departments

- Promote collaboration across teams.



## Deming and his 14 points for management

10. Eliminate slogans, exhortations, and targets for the workforce
  - Avoid meaningless numerical targets; focus on improving the process.
11. Eliminate numerical quotas and management by objectives
  - Avoid focusing solely on numbers; focus on quality and continuous improvement.
12. Remove barriers to pride of workmanship
  - Employees should be able to take pride in their work.
13. Institute education and self-improvement
  - Promote ongoing learning for everyone in the organization.
14. Put everyone in the company to work to accomplish the transformation
  - Quality improvement is a shared responsibility at all levels.

## What is the use of Deming's 14 points?

- Deming's philosophy provides **principles and guidelines** that influence how a Quality Management Plan is **designed, executed, and improved**. In essence, his points ensure the QMP is **process-focused, preventive, and continuous**.
- Deming's 14 Points **guide the structure and philosophy of a Quality Management Plan** by:
  - Focusing on process improvement, not just inspection.
  - Encouraging training, leadership, and collaboration.
  - Promoting preventive and continuous quality management.

# Juran and the important of top management commitment to Quality

## 1. Build Awareness of the Need for Improvement

Management must communicate the importance of quality and potential benefits.

## 2. Set Goals for Improvement

Define clear, measurable quality objectives.

## 3. Organise to Reach the Goals

Assign responsibility and structure teams for improvement projects.

## 4. Provide Resources

Ensure adequate budget, tools, training, and personnel.

## 5. Train Personnel

Equip staff with necessary skills for quality improvement.

# Juran and the important of top management commitment to Quality

## 6. Take Action to Improve

Implement process changes or corrective actions based on analysis.

## 7. Report Progress

Track and communicate improvement efforts using metrics.

## 8. Give Recognition

Acknowledge teams and individuals contributing to quality improvement.

## 9. Communicate Results

Share successes, failures, and lessons learned organization-wide.

## 10. Maintain Momentum

Integrate improvements into standard processes and continue seeking enhancements.

## How does Juran's 10 points relate to QMP?

- A Quality Management Plan is a document that defines how quality will be managed, monitored, and improved throughout a project. Juran's steps provide a structured framework for ensuring that the plan is actionable, systematic, and management-led.
- Juran's 10 Steps provide a practical roadmap for embedding quality into the Quality Management Plan. They ensure that:
  - Quality objectives are clear and measurable.
  - Processes are well-defined and resourced.
  - Improvement is continuous and supported by management.

# Cost of Quality & Trade-offs in IT Projects

- The **Cost of Quality** is the total cost incurred to ensure a product or service meets quality requirements. In other words, it's **the cost of preventing, detecting, and correcting defects**.
- Quality management requires balancing **time, cost, scope, and quality**. Here's how trade-offs work:
  - Cost vs Quality
    - Higher quality often requires **more investment** (prevention, testing, training). Cutting costs can increase defects and rework.
  - Time vs Quality
    - Tight schedules may force **reduced testing or shortcuts**, risking quality. Extending timelines allows thorough QA but may increase costs.
  - Scope vs Quality
    - Adding more features can dilute focus on **quality of existing functionality**.
  - Prevention vs Quality
    - Investing in **prevention (training, standards)** reduces defects later.

# Cost of Conformance and Cost of Non-Conformance

| Aspect     | Cost of Conformance                                                               | Cost of Non-Conformance                                                    |
|------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Definition | The cost of <b>ensuring that a product or service meets quality requirements.</b> | The cost incurred when the product <b>fails to meet quality standards.</b> |
| Focus      | Prevent defects proactively                                                       | Reactively fix defects                                                     |
| Components | Prevention + Appraisal                                                            | Internal + External Failures                                               |
| IT Example | Automated tests, code reviews, training                                           | Bug fixes, downtime, customer complaints                                   |
| Goal       | Minimize future costs and improve reliability                                     | Minimize impact of failures                                                |

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# For You!

## End of Lecture Questions

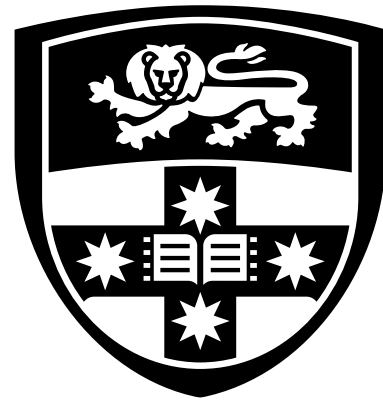
- Define **quality** in the context of IT project management.
- Explain the difference between **quality assurance (QA)** and **quality control (QC)**.
- Distinguish between **Cost of Conformance (CoC)** and **Cost of Non-Conformance (CoNC)**.
- Explain the **80/20 rule (Pareto Principle)** and its relevance to quality management.
- List at least **five tools and techniques** used in quality management and briefly describe their purpose.

## Case Study

- TechSolutions Pty Ltd is developing a **cloud-based Human Resource Management (HRM) system** for a large client. The system includes modules for **Employee Management, Payroll, Attendance, and Performance Tracking**. The project is under a tight 6-month deadline.
- During the first major release (end of Month 4), the client reports multiple defects:
  - Payroll module miscalculates overtime for certain employees.
  - Attendance module occasionally fails to sync with biometric devices.
  - Performance module crashes when generating reports for more than 500 employees.
- The QA team had conducted **unit and integration testing**, but due to time constraints, **system and user acceptance testing were limited**.
- The project manager is concerned about **high defect costs, delayed delivery, and potential client dissatisfaction**.

## Case study based questions

- Identify the **types of testing** that were missed and explain how they could have prevented these defects.
- Using **Cause-and-Effect (Ishikawa) analysis**, suggest potential root causes for the Payroll module defect.
- Categorize the defects reported into **Cost of Conformance (CoC)** and **Cost of Non-Conformance (CoNC)**.
- Propose **prevention measures** for future releases using **Juran's 10 Steps to Quality Improvement**.
- How can **Deming's 14 Points for Management** be applied to improve the HRM system's development process?
- Discuss the role of **top management commitment** in preventing similar issues in future projects.



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